Automatic Rebar Detailing System

Engineering Design Report

Project Details

Project Name: Slab Fig 4 Audit
Designer Name: Engineer
Date: 2026-09-08
Remarks: IS 456 Fig. 4 approximate digitization audit

INPUTS

Length: 5 m
Width: 3 m
Slab Thickness: 150 mm
Support/Wall Thickness: 150 mm
Clear Cover: 25 mm
Floor Finish Load: 1.5 kN/m²
Live Load: 3 kN/m²
Concrete Grade: M35
Steel Grade: Fe500
Support Type: Simply Supported
Unit Weight of Concrete: 25 kN/m³
Preferred Bar Diameter: 12 mm

LOAD CALCULATION

Slab thickness: 150 mm
Self weight: 3.75 kN/m²
Floor finish: 1.5 kN/m²
Dead load: 5.25 kN/m²
Live load: 3 kN/m²
Service load: 8.25 kN/m²
Ultimate load: 12.375 kN/m²

GEOMETRY

Clear span: 2.85 m
Effective short span: 3 m
Effective long span: 5 m
Aspect ratio ly/lx: 1.667
Slab classification: Two Way Slab
Effective span method: IS 456:2000 Clause 22.2: effective span = clear span + support width; for the provided span input, clear span = total span - support width and effective span = total span. Short and long directions calculated independently.

TWO-WAY MOMENT DESIGN

Coefficient short: 0.0497
Coefficient long: 0.0337
Mu short: 5.535 kN·m/m
Mu long: 3.753 kN·m/m
Moment equation: Mx = alphaX x w x lx^2; My = alphaY x w x lx^2 (IS 456 Table 26 coefficient convention).
Short span used for both moments, lx: 3 m
Coefficient source: IS 456:2000 Table 26 with linear interpolation between m = 1.6 and 1.7.

FLEXURAL DESIGN

SHORT DIRECTION
b: 1000 mm
d: 121 mm
Mu: 5.535 kN·m/m
Ast required: 180 mm²

Ast minimum: 180 mm²
Ast provided: 182.78 mm²
Bar arrangement: 8 mm @ 275 mm c/c
xu: 6.31 mm
xu,max: 55.66 mm
Moment capacity: 9.41 kN·m/m
Status: PASS

LONG DIRECTION
b: 1000 mm
d: 113 mm
Mu: 3.753 kN·m/m
Ast required: 180 mm²
Ast minimum: 180 mm²
Ast provided: 182.78 mm²
Bar arrangement: 8 mm @ 275 mm c/c
xu: 6.31 mm
xu,max: 51.98 mm
Moment capacity: 8.77 kN·m/m
Status: PASS

SHEAR

Critical section: At d = 121 mm from the support face; one-metre strip reaction = 18.563 kN/m.
Vu: 17.065 kN/m
tau_v: 0.141 N/mm²
tau_c: 0.3008 N/mm²
tau_c,max: 3.7 N/mm²
Tension steel percentage for Table 19: 0.1511 %
Concrete shear capacity: 36.402 kN/m
Reinforcement: Not required / not applicable because concrete shear capacity is adequate.
Status: PASS

DEFLECTION

Governing direction: Long direction
Governing span: 5 m
Governing effective depth: 113 mm
Basic L/d: 20
Tension steel percentage: 0.1618 %
Ast required: 180 mm²/m
Ast provided: 182.78 mm²/m
Service steel stress fs: 285.59 N/mm²
pt bracket: 0.15 to 0.25 %
fs bracket: 250 to 300 N/mm²
pt interpolation ratio: 0.117522
fs interpolation ratio: 0.711785
Clamping: NO
Modification factor: 1.066737
Actual L/d: 44.25
Permissible L/d: 21.33
Method: IS 456:2000 Clause 23.2.1 basic span/depth ratio multiplied by the Fig. 4 modification factor.
Source: IS 456:2000 Clause 23.2.1(c), Fig. 4; values are approximate digitizations of the published graph image at https://www.civilconcept.com/wp-content/uploads/2022/02/Modification-factor-figure-IS-Code-456-2000.jpg.
Dataset version: fig4-approx-2026-09
Interpolation: Bilinear interpolation between the explicitly named fs curves and shared pt points. Inputs outside the configured range are clamped and reported.
Status: FAIL

DETAILING

Short reinforcement: 8 mm @ 275 mm c/c

Long reinforcement: 8 mm @ 275 mm c/c
Short spacing: 275 mm
Long spacing: 275 mm
Short maximum spacing: 300 mm
Long maximum spacing: 300 mm
Short spacing status: PASS
Long spacing status: PASS
Short clear spacing: 267 mm
Long clear spacing: 267 mm
Minimum clear spacing: 25 mm
Clear-spacing rule: IS 456:2000 Clause 26.3.2: adopted minimum clear spacing is 25 mm for this design input set; the larger of bar diameter and 5 mm above nominal maximum aggregate size governs when aggregate size is supplied.
Maximum-spacing rule: IS 456:2000 Clause 26.3.3: main reinforcement spacing shall not exceed 3d or 300 mm, whichever is less. Both directions of a two-way slab are treated as main flexural reinforcement.
Short spacing check: PASS
Long spacing check: PASS
Cover: 25 mm
Corner restraint: FREE_TO_LIFT
Corner torsion reinforcement: Not required under the explicit FREE_TO_LIFT assumption.
Corner torsion status: Not required

FINAL CHECKS

Effective span validity: PASS | Effective span is finite and positive.
Slab classification: PASS | Aspect ratio = 1.667 -> Two Way Slab. | limit 2
Short-direction flexural capacity: PASS | MuProvided 9.41 >= required 5.535 and xu <= xu,max. | limit 5.535
Long-direction flexural capacity: PASS | MuProvided 8.77 >= required 3.753 and xu <= xu,max. | limit 3.753
Short required reinforcement: PASS | AstProvided 182.78 >= AstRequired 180. | limit 180
Long required reinforcement: PASS | AstProvided 182.78 >= AstRequired 180. | limit 180
Short minimum reinforcement: PASS | AstProvided 182.78 >= AstMin 180. | limit 180
Long minimum reinforcement: PASS | AstProvided 182.78 >= AstMin 180. | limit 180
Short maximum spacing: PASS | Provided spacing 275 <= max 300. | limit 300
Long maximum spacing: PASS | Provided spacing 275 <= max 300. | limit 300
Short clear spacing: PASS | Clear spacing 267 >= minimum 25 mm. | limit 25
Long clear spacing: PASS | Clear spacing 267 >= minimum 25 mm. | limit 25
Shear check: PASS | Shear stress is within concrete shear capacity. | limit 0.3008
Deflection control: FAIL | Actual L/d exceeds the permissible modified L/d; slab thickness/depth is insufficient for deflection control. | limit 21.33
Cover and detailing: PASS | Nominal cover 25 mm is valid, slab thickness 150 mm. | limit 150
Corner torsion restraint: PASS | Corners are explicitly configured as free to lift; corner torsion reinforcement is not required under this assumption. | limit 0
Final status: NOT APPROVED